Note 7: Pulse Modulation

ELEG2310B Principles of Communication Systems

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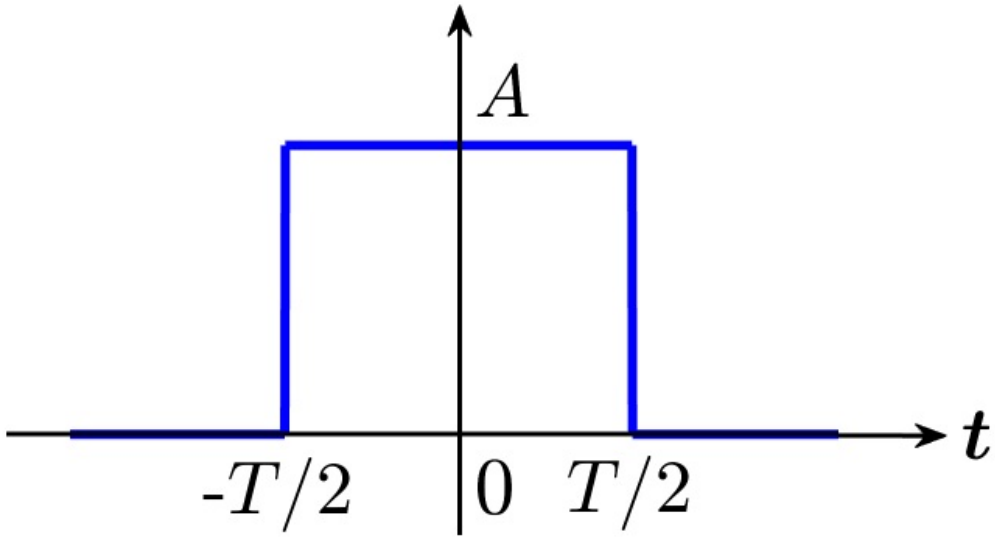
Carrier

Carrier Modulation: AM, DSB-SC, SSB, QAM, PM and FM

- Carrier refers to a continuous sinusoidal wave

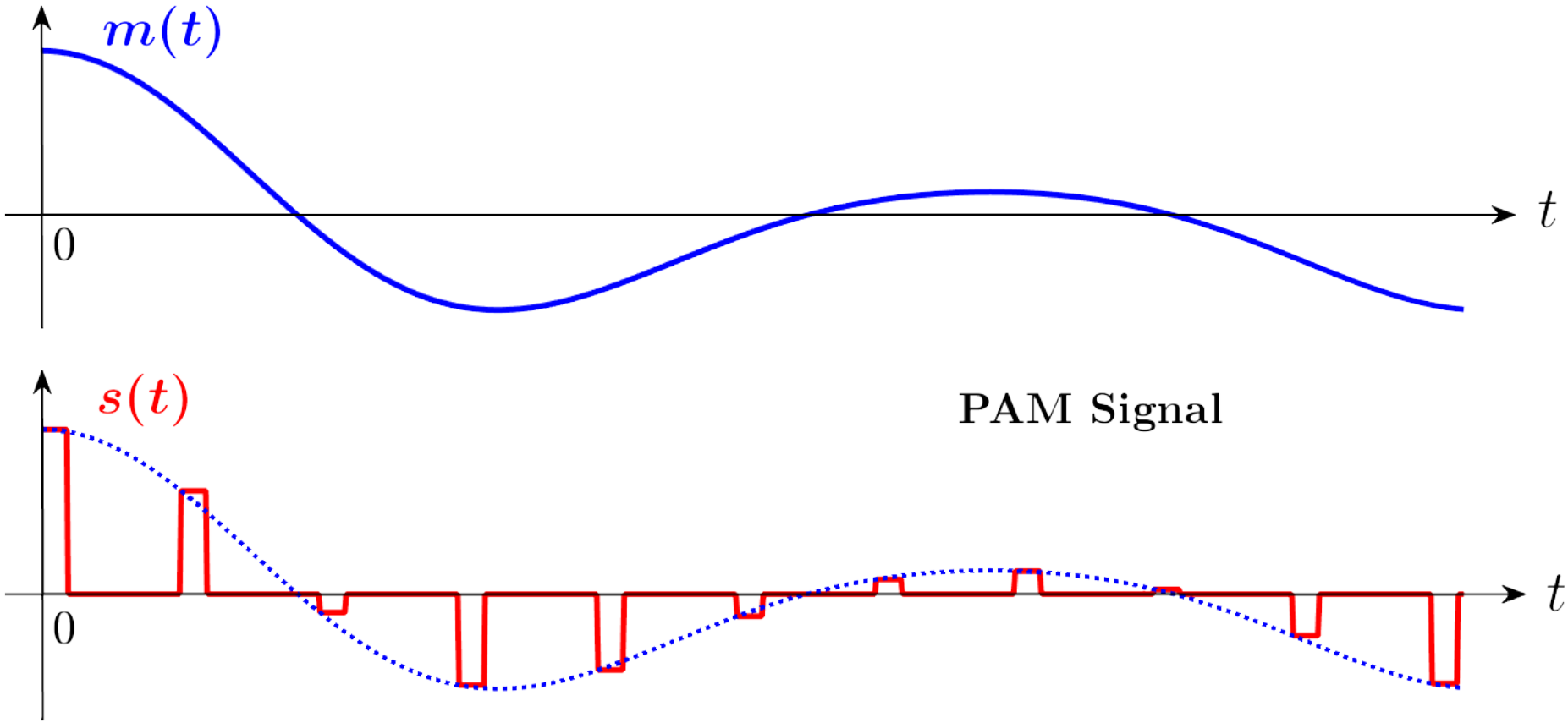
- In communication, when we say carrier, usually imply sinusoidal carrier

We could use other carriers to carry information, one alternative is **pulse**.



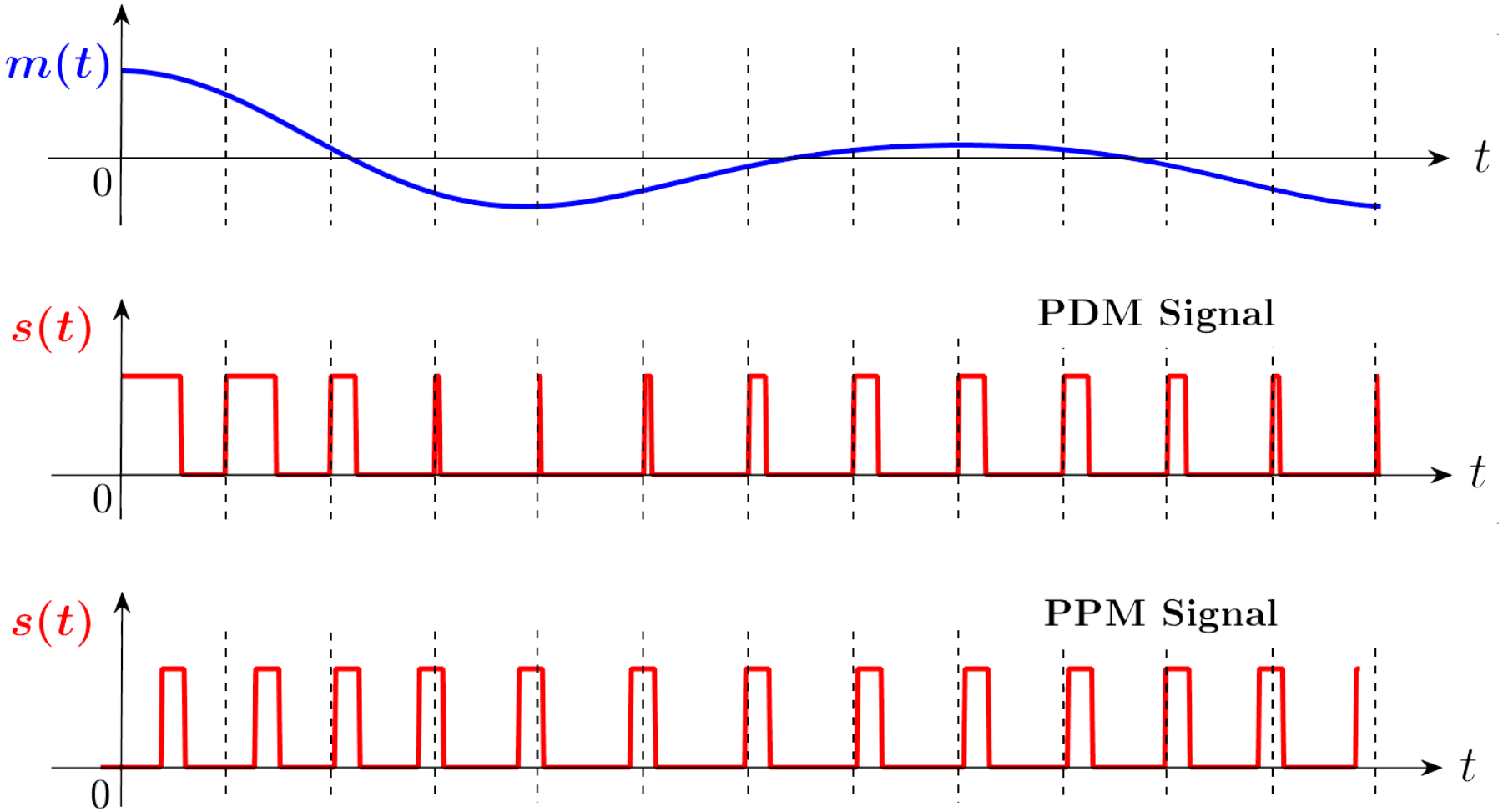
Pulse Amplitude Modulation

Pulse Amplitude Modulation (PAM): Use **pulse amplitude** to carry information.



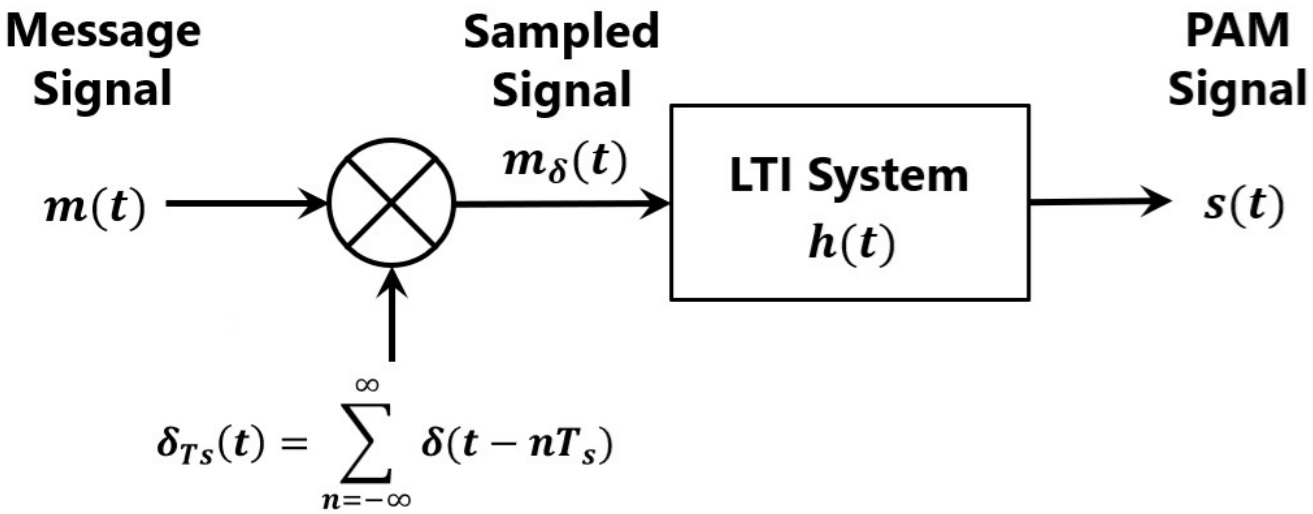
PDM and PPM

Pulse Duration Modulation (PDM): Use **pulse duration/width** to carry information. **Pulse Position Modulation (PPM)**: Use **pulse position** to carry information.



PAM

Sample-and-hold

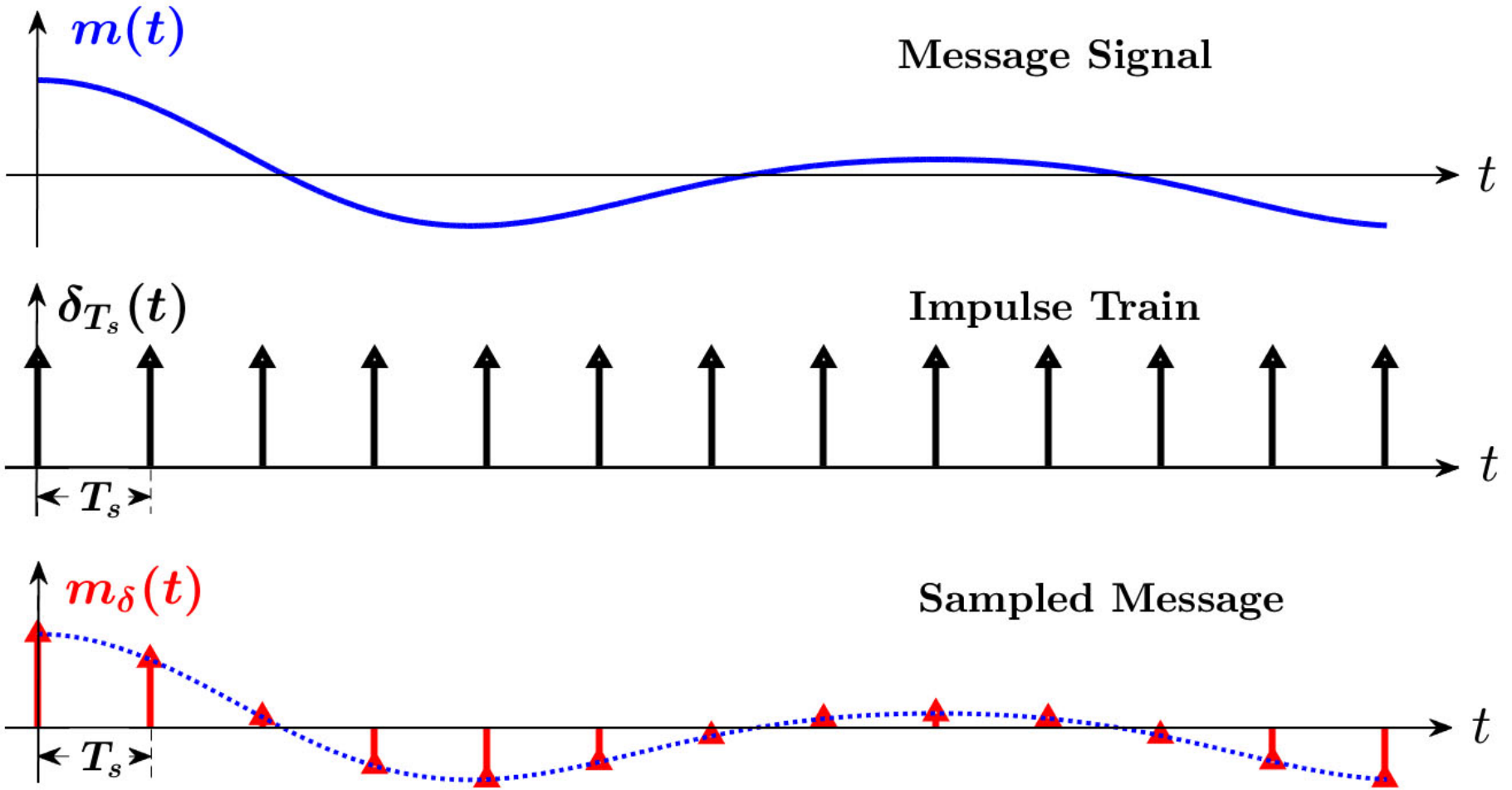


- Samples $m(t)$ for every T_s seconds, resulting a sequence of **samples**;
- each sample $m(nT_s)$ is carried by a **pulse $h(t)$** with duration T with $T < T_s$, where $h(t) = \begin{cases} 1\ , & 0 < t < T \\ 0\ , & \text{otherwise} \end{cases}$

Sampling Theorem

An ideal sampled signal is

$$m_\delta(t) = m(t)\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} m(nT_s)\delta(t - nT_s)$$



Sampling Theorem (Cont')

$\delta_{T_s}(t)$ is an impulse train with period T_s, its Fourier series is

$$\delta_{T_s}(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi kt}{T_s}}, \text{ where } a_k = \frac{1}{T_s}, \ \forall k.$$

Then, $m_\delta(t) = m(t)\delta_{T_s}(t) = \frac{1}{T_s}\sum_{k=-\infty}^{\infty} m(t)e^{j\frac{2\pi kt}{T_s}}$

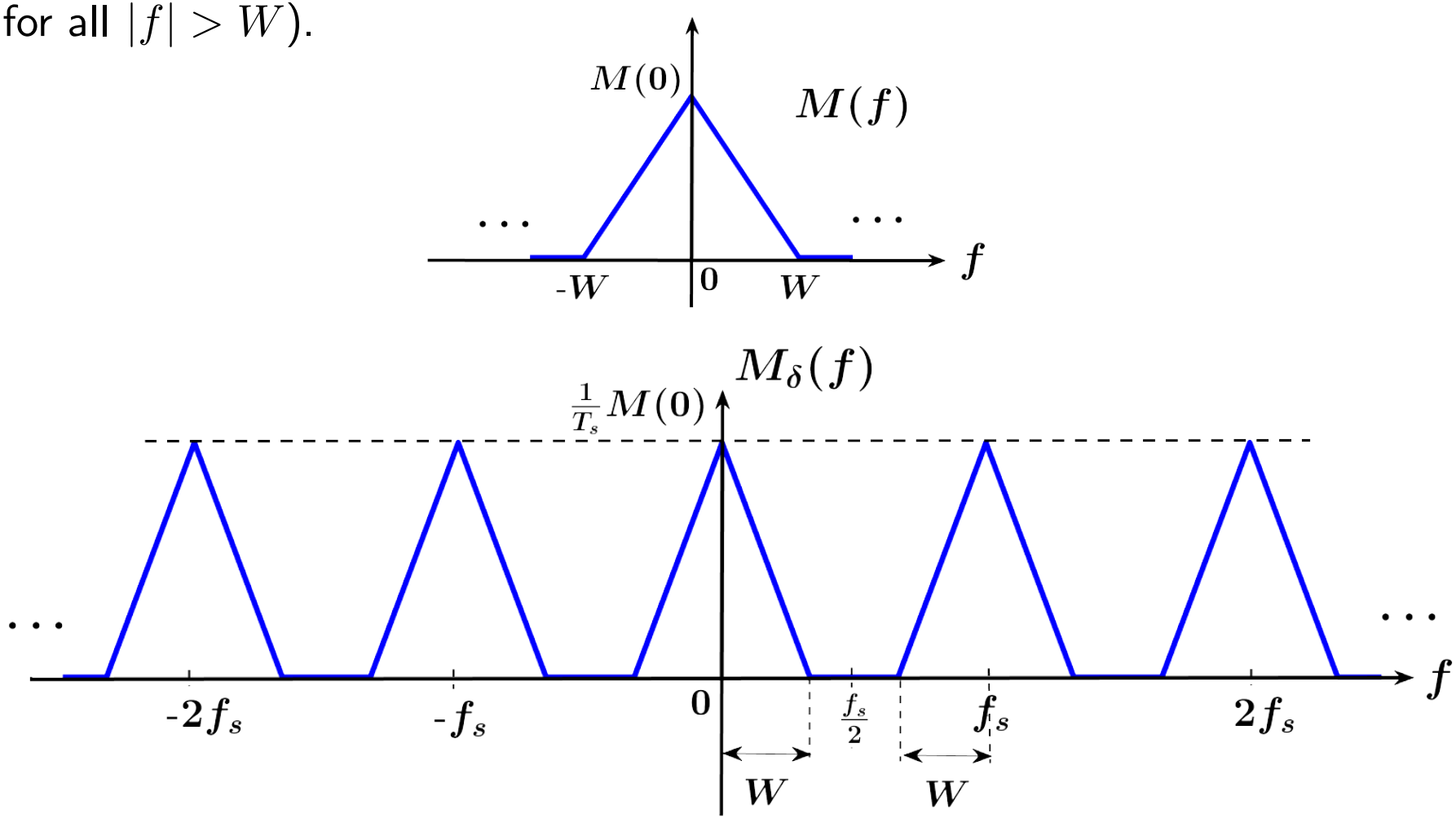
The spectrum of sampled signal $m_\delta(t)$ is

$$\boldsymbol{M_\delta(f)} = \frac{1}{T_s}\sum_{k=-\infty}^{\infty} \mathcal{F}\{m(t)e^{j\frac{2\pi kt}{T_s}}\}$$

$$= \boldsymbol{\frac{1}{T_s}\sum_{k=-\infty}^{\infty} M(f - kf_s)}, \text{ where } f_s = \frac{1}{T_s} \text{ is the sampling frequency}$$

Sampling Theorem (Cont')

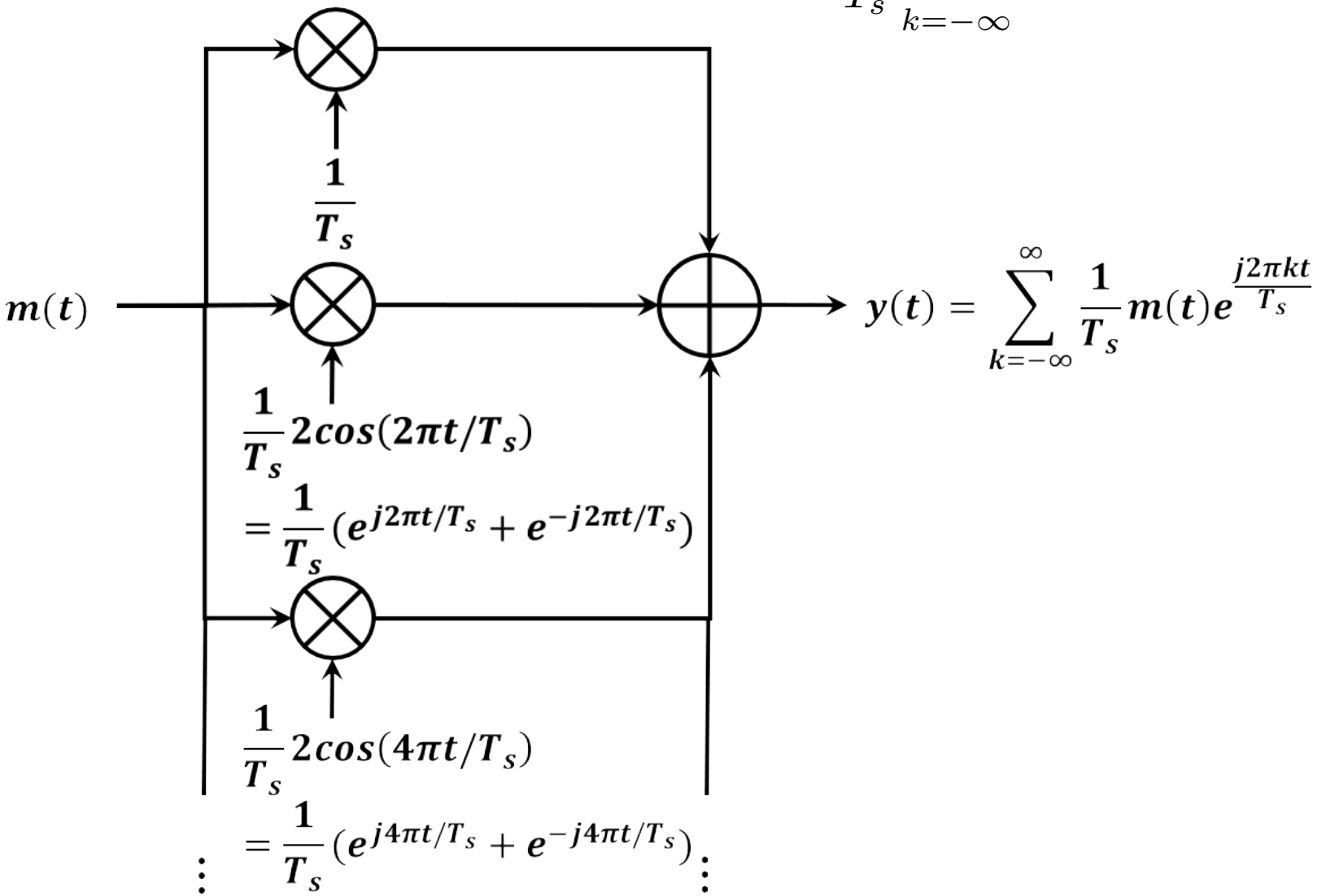
- Signal $m(t)$ is strictly bandlimited with bandwidth W Hz (i.e., $M(f) = 0$, for all $|f| > W$).



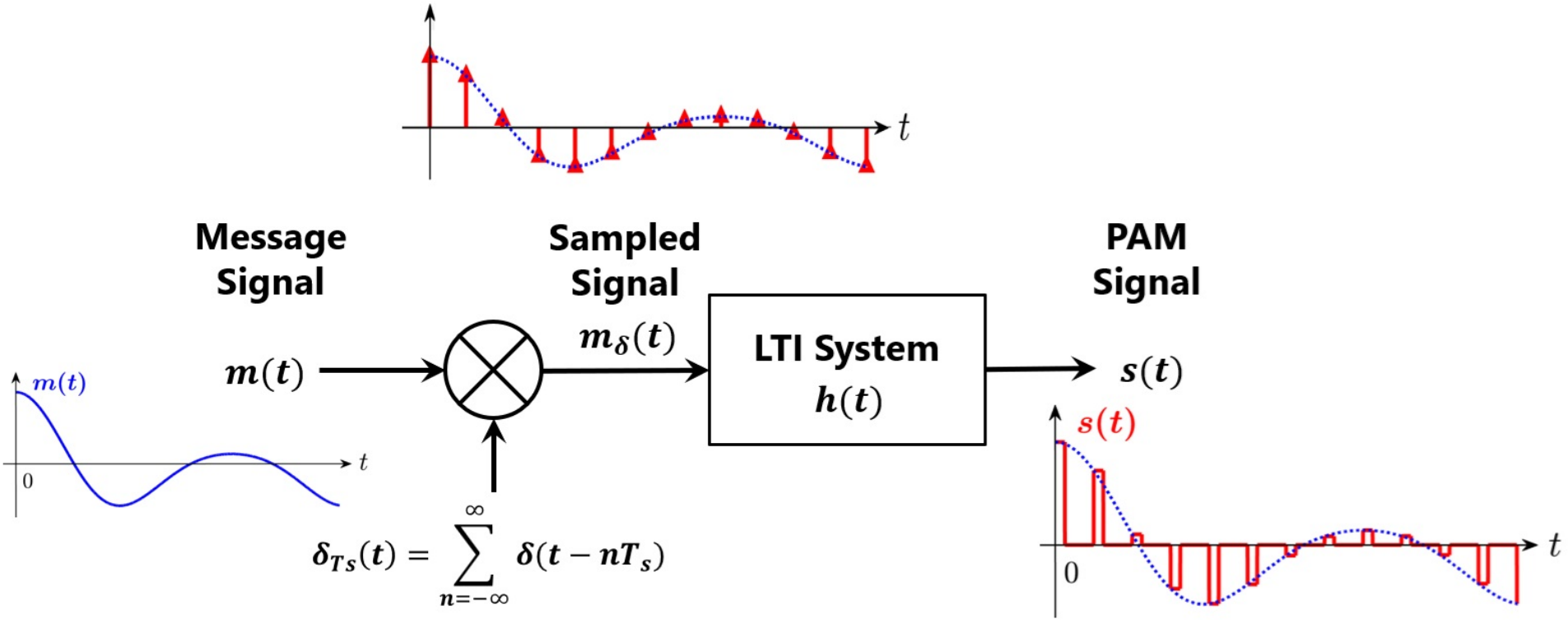
- $W \leq \dfrac{f_s}{2} \Rightarrow \boldsymbol{f_s \geq 2W}$, where $\boldsymbol{2W}$ is the well known **Nyquist Rate**.

Sampling Theorem (Cont')

Alternate way to obtain sampled signal: $m_\delta(t) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} m(t) e^{j\frac{2\pi k t}{T_s}}$



PAM Signals

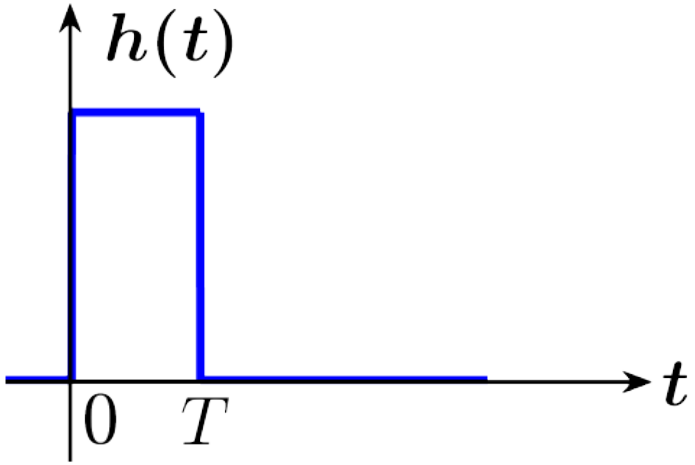


$$m_\delta(t) = m(t)\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} m(nT_s)\delta(t - nT_s)$$

PAM Signal: $s(t) = \int_{-\infty}^{\infty} h(\tau) m_\delta(t-\tau)\,\mathrm{d}\tau$, where $h(t) = \begin{cases} 1\ , & 0 < t < T \\ 0\ , & \text{otherwise} \end{cases}$

PAM Signals (Cont')

The impulse response of the LTI system is $h(t) = \begin{cases} 1\ , & 0 < t < T \\ 0\ , & \text{otherwise} \end{cases}$



i.e., $h(t) = \text{rect}\left(\frac{t - T/2}{T}\right)$

So, $H(f) = Te^{-j\pi fT}\text{sinc}(fT)$.

PAM Signals (Cont')

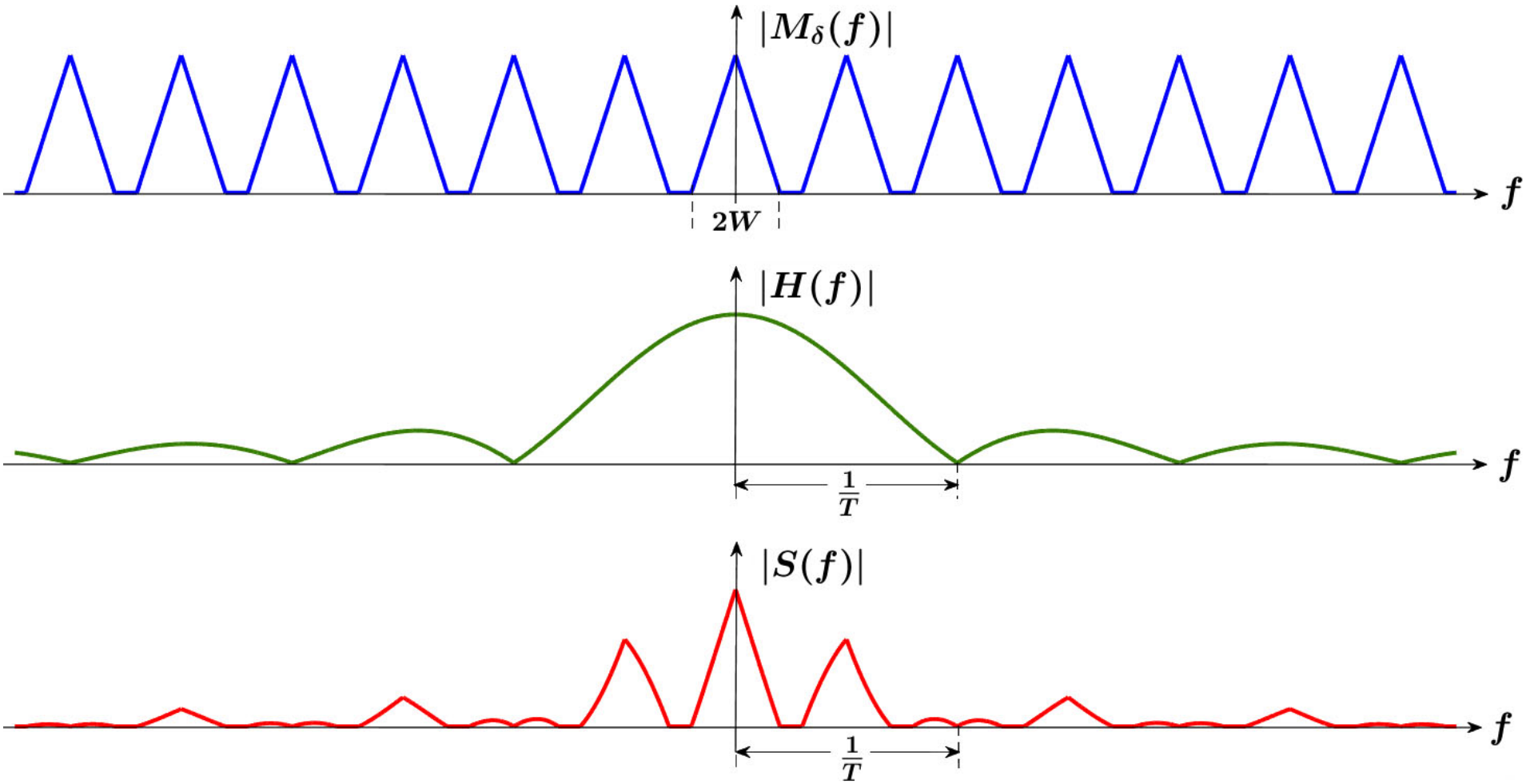
PAM Signal: $s(t) = m_\delta(t) * h(t)$

With $M_\delta(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} M(f - kf_s)$ and $H(f) = Te^{-j\pi fT}\text{sinc}(fT)$,

the spectrum of PAM signal is

$$S(f) = M_\delta(f)H(f) = \left[\frac{1}{T_s} \sum_{k=-\infty}^{\infty} M(f - kf_s)\right] H(f)$$

Spectrum of PAM Signals



PAM

Modulation

PAM signals are not strictly bandlimited.

PAM is not a bandwidth-efficient scheme.

Bandwidth $B_T = \frac{1}{T} \geq \frac{1}{T_s} \geq 2W$, which means PAM uses at least twice of the message bandwidth to transmit.

Demodulation

Assume the assumptions for perfect reconstruction in the sampling theorem hold,

- Detect the values of the sample $\left\{m(nT_s)\right\}_{n=-\infty}^{\infty}$ from the PAM signal $s(t)$ via sampling;
- Apply a reconstruction interpolation formula.

Signal Recovery / Reconstruction

Two assumptions:

- $m(t)$ is strictly bandlimited with bandwidth W Hz
- the sampling frequency $f_s \geq 2W$

$m(t)$ can be retrieved by applying **an ideal lowpass filter** with cutoff frequency $\boldsymbol{\frac{f_s}{2}}$ on $m_\delta(t)$.

The frequency response of the lowpass filter is

$$\boldsymbol{H_{LP}(f) = \begin{cases} 1\ , & -\frac{f_s}{2} < f < \frac{f_s}{2} \\ 0\ , & \textbf{otherwise} \end{cases}}$$

And, the corresponding impulse response is $\boldsymbol{h_{LP}(t) = f_s \mathrm{sinc}(f_s t)}$.

Then, we have $\boldsymbol{M_\delta(f) H_{LP}(f) = \frac{1}{T_s} M(f)}$,

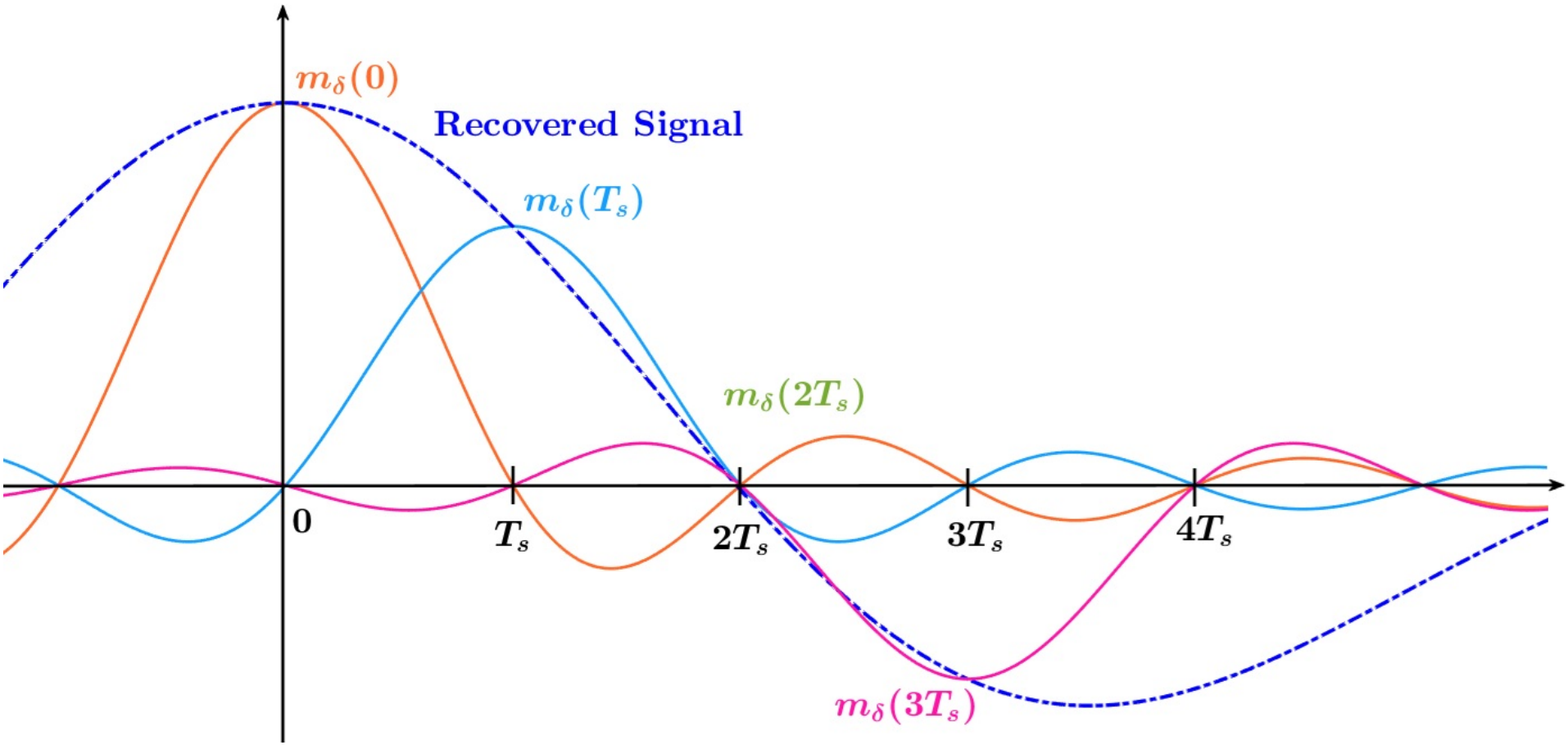
and $\boldsymbol{\int_{-\infty}^{\infty} h_{LP}(\tau) m_\delta(t-\tau)\,\mathrm{d}\tau = \frac{1}{T_s} m(t)}$.

Signal Recovery (Cont')

$$
\begin{aligned}
g(t) &= T_s \int_{-\infty}^{\infty} h_{LP}(\tau) m_\delta(t-\tau)\,\mathrm{d}\tau \\
&= T_s \int_{-\infty}^{\infty} h_{LP}(\tau) \Big[\sum_{n=-\infty}^{\infty} m(nT_s)\delta(t-\tau-nT_s) \Big]\,\mathrm{d}\tau \\
&= T_s \sum_{n=-\infty}^{\infty} m(nT_s) \int_{-\infty}^{\infty} h_{LP}(\tau)\delta(t-\tau-nT_s)\,\mathrm{d}\tau \\
&= T_s \sum_{n=-\infty}^{\infty} m(nT_s) h_{LP}(t-nT_s) \\
&= T_s \sum_{n=-\infty}^{\infty} m(nT_s) \big[f_s \mathrm{sinc}(f_s(t-nT_s)) \big] \\
&= \sum_{n=-\infty}^{\infty} \boldsymbol{m(nT_s)\mathrm{sinc}\Big(\frac{t-nT_s}{T_s}\Big)}
\end{aligned}
$$

It is known as the **Interpolation Formula**.

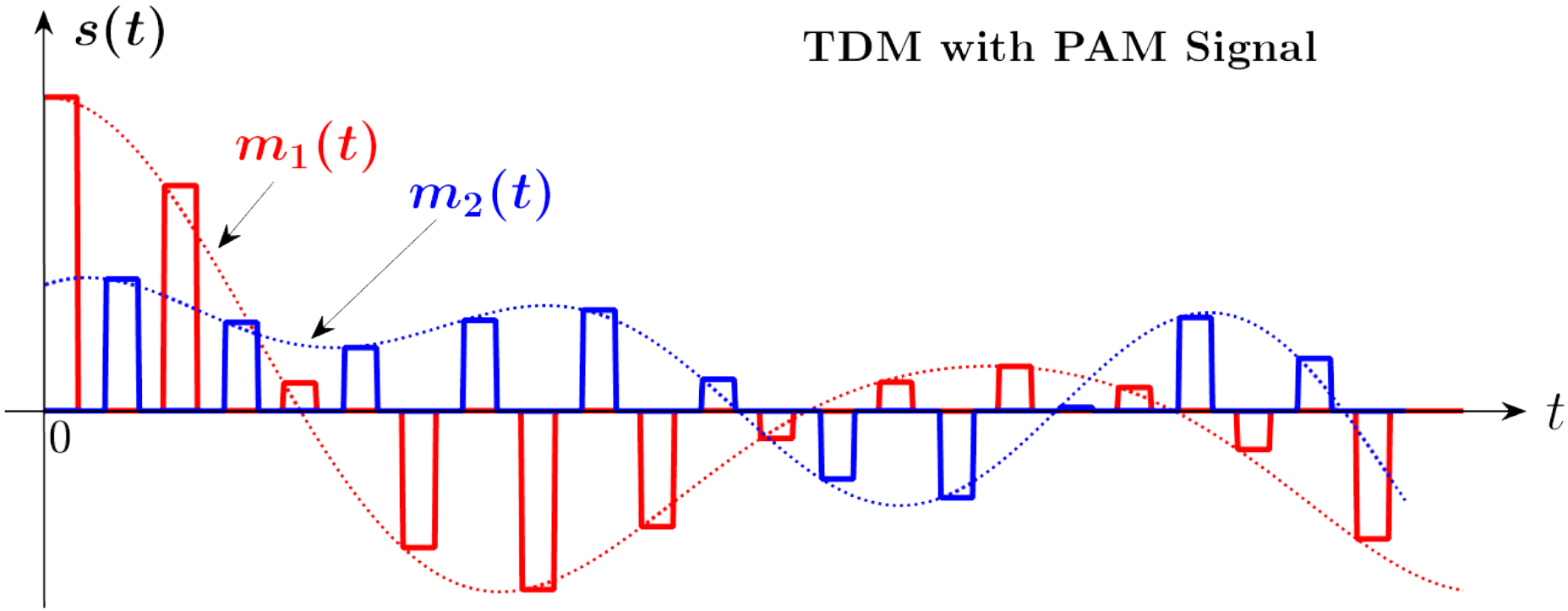
Interpolation Formula



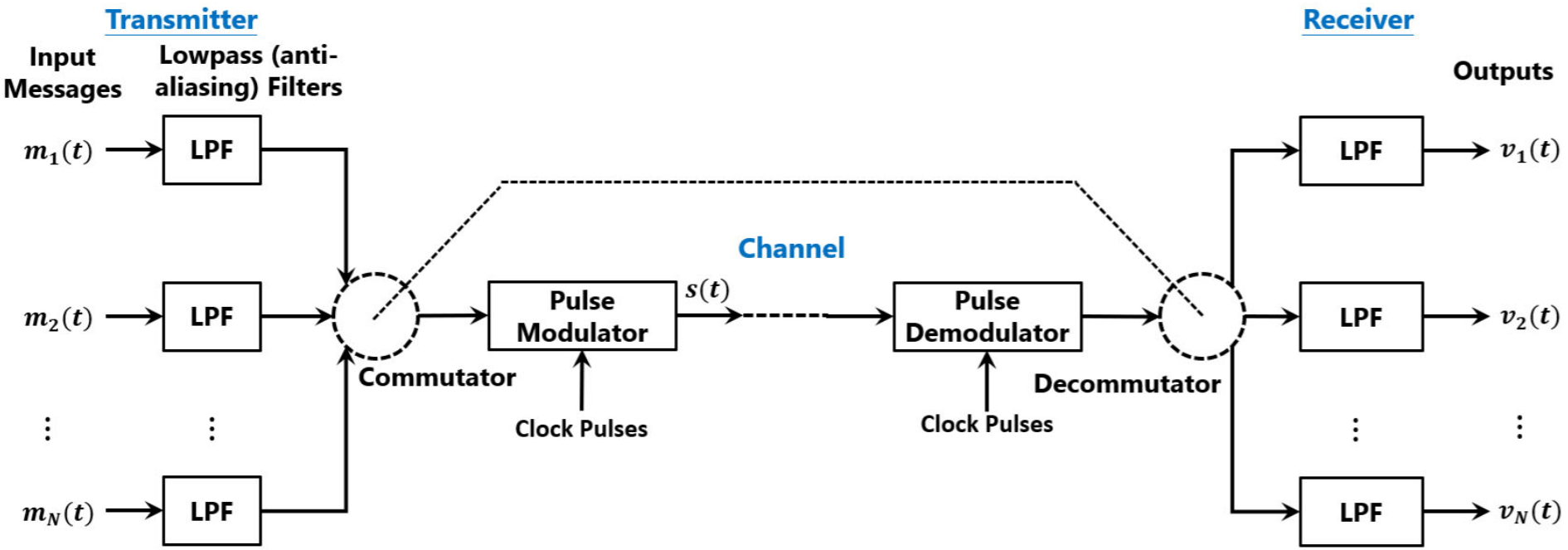
Time Sharing

Time Division Multiplexing (TDM) aims to multiplex multiple message signals within one modulated signal by sharing the channel in time-interweaving manner.

Example: TDM signal of two message signals



TDM System



Commutator

- takes N samples, each from one message signal within the period T_s;
- interweaves the N samples for PAM transmission.

Decommutator

- does the inverse operation of the commutator to demultiplex the samples.

Commutator and decommutator must maintain timing synchronism in order for the system to work properly.